ML Assignment 1

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Part 1:

Q.1 Given: $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

(i) AB

$$= \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 3(1)+4(0) & 3(2)+4(-1) \\ 2(1)+1(0) & 2(2)+1(-1) \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

(ii) $BA = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1(3)+2(2) & 1(4)+2(1) \\ 0(3)+(-1)(2) & 0(4)+(-1)(1) \end{bmatrix}$

$$\therefore BA = \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$$

(iii) $(A+B)^T$:

$$A+B = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}$$

$$\therefore (A+B)^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$$

(iv) AB^T: $B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$

$$\Rightarrow A.B^T = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 3(1)+4(2) & 3(0)+4(-1) \\ 2(1)+1(2) & 2(0)+1(-1) \end{bmatrix}$$

$$\therefore A.B^T = \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}$$

(v) trace(A):

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$$

$\Rightarrow$ trace(A) = 3 + 1

$\therefore$ trace(A) = 4

(vi) trace(AB):

$$AB = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$ — <From (i)>

$\Rightarrow$ trace(AB) = 3 + 3

$\therefore$ trace(AB) = 6

Q2] Given: $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

(i) Characteristic eq^n: $M - \lambda I = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

$\Rightarrow M - \lambda I = \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}$ — (a) $\Rightarrow$ characteristic matrix

For Eigenvalues: Solve characteristic eq^n:

$$|M - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow (2-\lambda)^2 - (-1)(-1) = 0$$

$\therefore 4 - 4\lambda + \lambda^2 - 1 = 0$

Rearranging: $\lambda^2 - 4\lambda + 3 = 0$

$\Rightarrow \lambda^2 - \lambda - 3\lambda + 3 = 0 \Rightarrow \lambda(\lambda - 1) - 3(\lambda - 1) = 0$

$\Rightarrow (\lambda - 3)(\lambda - 1) = 0$

$\therefore \lambda = 3$ or $\lambda = 1 \Rightarrow$ Eigen values

For Eigenvectors,

Case (i): $\lambda = 3$

$(M - \lambda I)v = 0 \Rightarrow (M - 3I)v = 0$

$$\Rightarrow \left(\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} (-1)(v_1) + (-1)(v_2) \\ (-1)(v_1) + (-1)(v_2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow -v_1 - v_2 = 0 \Rightarrow v_2 = -v_1$

If $v_1 = 1$, then $v_2 = -1$

$$\therefore v_1 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Case (ii): $\lambda = 1$

$(M - \lambda I)v = 0 \Rightarrow (M - I)v = 0$

$$\Rightarrow \left\{ \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} v_1 + (-1)v_2 \\ (-1)v_1 + v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow v_1 - v_2 = 0$

$\Rightarrow v_1 = v_2$

If $v_1 = 1$, then $v_2 = 0$

$$\therefore v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(i) Verification: $(Mv_1 = \lambda_1 v_1)$

Case 1: When $\lambda = 3$

$$Mv_1 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2(1) + (-1)(-1) \\ (-1)(1) + 2(-1) \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \lambda_1 v_1$$

$\Rightarrow Mv_1 = \lambda_1 v_1$

Case 2: When $\lambda = 1$

$$Mv_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(1) + (-1)(1) \\ (-1)(1) + 2(1) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \lambda_2 v_2$$

$\Rightarrow Mv_2 = \lambda_2 v_2$

Q3] Let $A=\begin{bmatrix}5 & -2\\ 1 & 4\end{bmatrix}$, $B=\begin{bmatrix}3 & 7\\ -1 & 2\end{bmatrix}$

To prove: $tr(AB)=tr(BA)$

Proof:

$$AB=\begin{bmatrix}5 & -2\\ 1 & 4\end{bmatrix}\cdot\begin{bmatrix}3 & 7\\ -1 & 2\end{bmatrix}=\begin{bmatrix}5(3)+(-2)(-1) & 5(7)+(-2)(2)\\ 1(3)+4(-1) & 1(7)+4(2)\end{bmatrix}$$

$$\therefore AB=\begin{bmatrix}17 & 31\\ -1 & 15\end{bmatrix}$$

$$BA=\begin{bmatrix}3 & 7\\ 1 & 2\end{bmatrix}\cdot\begin{bmatrix}5 & -2\\ 1 & 4\end{bmatrix}=\begin{bmatrix}3(5)+7(1) & 3(-2)+7(4)\\ 1(5)+2(1) & 1(-2)+2(4)\end{bmatrix}$$

$$\therefore BA=\begin{bmatrix}22 & 22\\ -3 & 10\end{bmatrix}$$

Now, $trace(AB)=17+15=32$

$trace(BA)=22+10=32$

$\therefore trace(AB)=trace(BA)$

Hence Proved

Q4] Given $C=\begin{bmatrix}4 & 7\\ 2 & 6\end{bmatrix}$

$|C|=4(6)-7(2)$

$=10$

$\because |C|\neq 0 \Rightarrow$ Inverse exists

$\Rightarrow$

$$C^{-1}=\frac{1}{|C|}\begin{bmatrix}6 & -7\\ -2 & 4\end{bmatrix}$$

$$=\frac{1}{10}\begin{bmatrix}6 & -7\\ -2 & 4\end{bmatrix}=\begin{bmatrix}6/10 & -7/10\\ -2/10 & 4/10\end{bmatrix}$$

$$\therefore C^{-1}=\begin{bmatrix}3/5 & -7/10\\ -1/5 & 2/5\end{bmatrix}$$

Part 2:

Q1] Given: $P(X=k) = c.\frac{3^k}{k!}$, $k = 0, 1, 2, \ldots$

a) $\sum_{k=0}^{\infty} P(X=k) = 1$

$\Rightarrow c.\sum_{k=0}^{\infty} \frac{3^k}{k!} = 1 \quad \ldots (I)$

Now, acc. to the Taylor series expansion, $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$

Here, $x = 3 \Rightarrow \sum_{k=0}^{\infty} \frac{3^k}{k!} = e^3$

Put in (I)

$\therefore c.e^3 = 1 \Rightarrow$

$c = \frac{1}{e^3}$, where c: Normalisation constant

b) Expectation $E[X] = \sum_{k=1}^{\infty} k.P(X=k)$

$= \sum_{k=1}^{\infty} k.\frac{e^{-3}3^k}{k!} =$

Now, $\frac{k}{k!} = \frac{k}{k(k-1)!} = \frac{1}{(k-1)!}$

$\therefore E[X] = \sum_{k=1}^{\infty} \frac{e^{-3}3^k}{(k-1)!}$

Now, acc to Taylor series expansion: $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

$\Rightarrow E[X] = 3e^{-3}\sum_{k=1}^{\infty} \frac{3^{k-1}}{(k-1)!}$

$= 3e^{-3}.e^3$

$\therefore E[X] = 3.e^{-3+3} = 3.e^0$

$\therefore E[X] = 3$

Variance: $Var[X] = E[X^2] - \{E[X]\}^2$

For $\{E[X]\}^2$:

$\Rightarrow E[X(X-1)] = \sum_{k=0}^{\infty} k.(k-1)\frac{e^{-3}3^k}{k!}$

For $k=0$ & $k=1$, $E[X(X-1)]=0$

$\therefore E[X(X-1)] = \sum_{k=2}^{\infty} k(k-1).\frac{e^{-3}3^k}{k!}$

$= \sum_{k=2}^{\infty} \frac{e^{-3}.3^k}{(k-2)!}$ ⟵ $\because \frac{k(k-1)}{k!} = \frac{1}{(k-2)!}$

$\therefore E[X(X-1)] = 9e^{-3}\sum_{k=2}^{\infty}\frac{3^{k-2}}{(k-2)!}$

Let $s = k-2$

$\therefore E[X(X-1)] = 9e^{-3}\sum_{s=0}^{\infty}\frac{3^s}{s!}$

$= 9e^{-3}(e^3)$

$\therefore E[X(X-1)] = 9$

$E[X^2] = E[X(X-1)] + E[X] = 9+3 = 12$

$\therefore var(X) = E[X^2] - \{E[X]\}^2 = 12 - (3)^2 = 3$

$\therefore var(X) = 3$

Q2] Given $P(A)=0.5$, $P(B)=0.3$ & $P(A|B)=0.6$

a) $P(A\cap B) = P(A/B).P(B)$

$= (0.6).(0.3)$

$\therefore P(A\cap B) = 0.18$

b) $P(B/A) = \frac{P(A\cap B)}{P(A)} = \frac{0.18}{0.5}$

$\therefore P(B/A) = 0.36$

c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.3 - 0.18$

$\therefore P(A \cup B) = 0.62$

Q3] Given $P(\text{Heads}) = \theta$

$\therefore P(\text{Tails}) = 1 - \theta$

Given sequence: H, H, T, H, T

Heads count: 3

Tails count: 1

a) Likelihood function:

$L(\theta) = P(H) \cdot P(H) \cdot P(T) \cdot P(H) \cdot P(T)$

$= \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta)$

$= \theta^3 \cdot (1-\theta)^2$

Taking Log likelihood:

$L(\theta) = \ln[\theta^3 (1-\theta)^2]$

$= 3\ln(\theta) + 2\ln(1-\theta)$

Now, taking first derivative & setting it to 0

$\frac{dL}{d\theta} = \frac{3}{\theta} - \frac{2}{1-\theta} = 0$

$\therefore \frac{3}{\theta} = \frac{2}{1-\theta}$

$\Rightarrow 3(1-\theta) = 2\theta$

$\Rightarrow 5\theta = 3$

$\Rightarrow \theta = \frac{3}{5}$

$\therefore \theta = 0.6$

b) Given $P(\theta) = 2\theta$, for $0 \le \theta \le 1$

$P(\theta/\text{Data}) = \frac{L(\theta) \cdot P(\theta)}{P(\text{Data})}$

$\because P(\text{Data})$ doesn't depend on data

$\therefore P(\theta/\text{Data}) \propto L(\theta) \cdot P(\theta)$

$\therefore P(\theta/Data) \propto [\theta^3(1-\theta)^2] \cdot [2\theta]$

$\propto \theta^4(1-\theta)^2$

Let, $g(\theta) = \ln[\theta^4(1-\theta)^2]$

$\therefore g(\theta) = 4\ln\theta + 2\ln(1-\theta)$

Now, taking first derivative & setting it to 0

$\therefore \frac{dg}{d\theta} = \frac{4}{\theta} - \frac{2}{1-\theta} = 0$

$\therefore \frac{4}{\theta} = \frac{2}{1-\theta}$

$\Rightarrow 4(1-\theta) = 2\theta$

$\therefore \theta = \frac{4}{6}$

$= \frac{2}{3}$

$\therefore \theta_{MAP} = 0.667$

Q4] Given dataset $X = 1, 3, 5, 7$, $n = 4$

Gaussian distribution: $L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$

Taking log,

$$l(\mu, \sigma^2) = \frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\left[\sum_{i=1}^{n}(x_i-\mu)^2\right] \quad \text{--- (I)}$$

$\frac{\partial l}{\partial \mu} = 0 - 0 + \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$ — Deriving I wrt μ

$\Rightarrow \sum_{i=1}^{n} x_i - n\mu = 0$

$\therefore \mu_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{4}[1+3+5+7] = \frac{16}{4} = 4$

$\therefore \mu_{MLE} = 4$

Now for σ^2_{MLE}

$$\frac{\partial L}{\partial \sigma^2} = \frac{-n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Setting it to 0,

$$\frac{-n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\cdot\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\Rightarrow -n\sigma^2 + \sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\therefore n\sigma^2 = \sum_{i=1}^{n}(x_i-\mu)^2$$

$$\Rightarrow 4\sigma^2 = \sum_{i=1}^{4}(x_i-9)^2$$

$$= \left[(1-9)^2 + (3-9)^2 + (5-9)^2 + (7-9)^2\right]$$

$$\Rightarrow 4\sigma^2 = 20$$

$$\therefore \sigma^2 = 5$$

Q.5] Given $\sigma^2 = 4$

$$\mu \sim N(0,1)$$

$$x = 5$$

$$P(x/\mu) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$= \frac{1}{\sqrt{8\pi}}\exp\left(-\frac{(5-\mu)^2}{8}\right)$$

Prior on μ is $\mu \sim N(\mu_0, \sigma_0^2) = N(0,1)$:

$$\therefore P(\mu) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{(\mu-0)^2}{2(1)}\right)$$

$$= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{\mu^2}{2}\right)$$

Now, by Baye's Theorem,

$$P(\mu/x) \propto P(x/\mu).P(\mu)$$

$$\therefore P(\mu/x) \propto \exp\left(-\frac{(5-\mu)^2}{8}\right) \cdot \exp\left(\frac{-\mu^2}{2}\right)$$

$$= \exp\left\{-\left[\frac{(5-\mu)^2}{8} + \frac{\mu^2}{2}\right]\right\}$$

Now, to maximise, introducing log-objective function $J(\mu)$:

$$J(\mu) = -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2} + c$$

Now, taking first derivative & setting it to 0

$$\frac{dJ}{d\mu} = -\frac{2(5-\mu)(-1)}{8} - \frac{2\mu}{2} = 0$$

$$\therefore \frac{5-\mu}{4} - \mu = 0$$

$$\Rightarrow 5 - \mu - 4\mu = 0$$

$$\therefore 5\mu = 5$$

$$\therefore \mu = 1$$

Q6] To prove: $\text{var}(X) = E(X^2) - [E(X)]^2$

Proof:

By defn. $\text{var}(X) = E[(X - E[X])^2]$

Now,

$$(X - E[X])^2 = X^2 - 2X \cdot E[X] + (E[X])^2$$

$$\Rightarrow \text{var}(X) = E[X^2 - 2X \cdot E[X] + (E[X])^2]$$

$$= E[X^2] - E[2X \cdot E[X]] + E[(E[X])^2]$$

$$= E[X^2] - E[2X \cdot E[X]] + (E[X])^2$$

Also, $E[2X \cdot E[X]] = 2E[X] \cdot E[X]$

$$= 2(E[X])^2$$

$\therefore$ $\text{var}(X) = E[X^2] - 2(E[X])^2 + (E[X])^2$

$\therefore$ $\text{var}(X) = E[X]^2 - (E[X])^2$

Hence proved

Part 3:

Q.1] To derive: $\text{var}(v^T X) = v^T \Sigma v$

Let $X \in R^d$ be a d-dimensional random vector.

$\Rightarrow$ Mean vector $\mu \in R^d$ is the expected value of X.

If $X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_d \end{bmatrix}$, then $\mu = E[X] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_d] \end{bmatrix}$

Let $v \in R^d$ be a deterministic unit vector specifying projection direction:

$$\|v\|_2 = 1 \Rightarrow v^T v = 1$$

Projection of X onto v is the dot product: $Y = v^T X$

Now,

Covariance Matrix $\Sigma = \text{Cov}(X) = E[(X-\mu)(X-\mu)^T]$

where, $(X-\mu)$: column vector of dimension $(d \times 1)$

$(X-\mu)^T$: row vector of dimension $(1 \times d)$

Now, $E[Y] = E[v^T X]$

$= v^T E[X]$

$\therefore E[Y] = v^T \mu$

Now,

$\text{var}(Y) = E[(Y - E[Y])^2]$, put $Y = v^T X$ & $E[Y] = v^T \mu$

$\therefore$ $Y - E[Y] = v^T X - v^T \mu = v^T (X - \mu)$

$\therefore$ $\text{Var}(v^T X) = E[(v^T (X-\mu)^2]$

For any real scalar z, $z^2 = z \cdot z^T$

Let $z = v^T (X-\mu) \Rightarrow z^T = (v^T (X-\mu))^T = (X-\mu)^T \cdot (v^T)^T = v (X-\mu^T)$

$\Rightarrow (v^T (X-\mu))^2 = (v^T (X-\mu))((X-\mu)^T v)$

$\Rightarrow (u^T(x-\mu))^2 = u^T(x-\mu)(x-\mu)^T u$

Put in $var(u^T X)$

$\Rightarrow var(u^T X) = E[u^T(x-\mu)(x-\mu)^T u]$

$= u^T E[(x-\mu)(x-\mu)^T]u$

But $\Sigma = E[(x-\mu)(x-\mu)^T]$

$\Rightarrow var(u^T X) = u^T \Sigma u$

Hence Proved

Defn) The covariance matrix (Σ) measures the pairwise linear dependence and individual variances among the elements X:

$$\Sigma = Cov(X) = E[(X-\mu)(X-\mu)^T]$$

Q2] # Code + visualisation

Q3] # Code + link to Q1

Q4] # Refer code as well

Trade-off between Explained variance and Dimensionality Reduction:

PCA aims to reduce the number of dimensions while preserving as much training data information as possible.

When reducing d dimensions to k dimensions, there is a trade-off:

- If k is small, dimensionality is greatly reduced, making process simpler and faster, but information may be lost.
- If k is large, more variance and data is preserved, but benefit of dimensionality reduction decreases.